

Tangible Immersive Authoring of Dynamic Mixed Reality Scenes with the World in Miniature Paradigm

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Figure 1: An overview of system workflow: (a) capture the environment and generate a miniature model of the environment; (b) select the desired model and adjust its scaling from the menu; (c) manipulate a physical cube to place it, supported by a spatio-temporal bounding volume; (d) check the instant results in the physical environment; (e) repeat for new models.

Abstract

We present work-in-progress on the development of a framework for immersive authoring of dynamic Mixed Reality scenes, employing the *World-in-Miniature* technique to reduce workload, using a tangible user interface for intuitive object placement, and incorporating spatio-temporal bounding volumes to avoid clipping with dynamic content. We developed a prototype on the Meta Quest 3 using its Passthrough camera access, and present the design and implementation details for such a system, along with insights that motivate promising directions for future work.

ACM Reference Format:

Xuyu Li and John Dingliana. 2025. Tangible Immersive Authoring of Dynamic Mixed Reality Scenes with the World in Miniature Paradigm. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771529>

1 Introduction

Situated mixed-reality (MR) experiences in application domains such as cultural heritage [Rubino et al. 2015] and theatre [Kim et al. 2023] necessitate embedding virtual content into specific locations in the real world. Immersive authoring [Lee et al. 2005], where this content is interactively positioned in situ, offers a more effective

approach than editing the scene indirectly using traditional desktop 3D modeling tools. However, such direct approaches require creators to physically move to different real-world locations to place objects, which can become cumbersome when the experience spans large spaces or remote locations. This is further exacerbated when dealing with dynamic content such as animated characters, which are ubiquitous in MR experiences but need to be carefully positioned to prevent unrealistic clipping against physical obstacles in the environment.

Prior works [Ihara and Kawaguchi 2022a,b] have shown that employing a World-in-Miniature (WIM) approach [Stoakley et al. 1995] for placing objects using gesture interaction can reduce physical and temporal demand in such tasks. Extending on this approach, we propose a novel framework that combines the WIM approach with a tangible user interface (TUI) [Ishii and Ullmer 1997] to allow users to intuitively place dynamic virtual assets in the real-world without need for a traditional mouse-keyboard interface but improving on the precision and ease-of-use of gestural interaction. Finally, spatio-temporal bounding volumes are used to avoid clipping animated sequences with real-world objects.

2 Design Goals

The objective of our system is to enable users to easily and efficiently author dynamic MR scenes at specific physical locations. Our target users include domain experts and designers who wish to quickly prototype personalized MR scenes without requiring significant deployment effort. To this end, we identified the following design goals: **efficient scene curation** (DG1): to let users quickly prototype comprehensive MR scenes, our system should support efficient curation of multiple digital models throughout

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SA Posters '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771529>

physical environments; **intuitive manipulation** (DG2): to reduce the complexity of model placement during the authoring process, our system should employ natural and intuitive interaction (i.e., moving and rotating) for manipulating the models; **direct authoring** (DG3): to improve the accessibility of the system and enable rapid design iteration, our system should allow users to prototype directly within the MR experience itself, without needing advanced setups or technical proficiency; and **clear visualization and feedback** (DG4): to enable effective design of dynamic MR scenes, our system should provide clear visualizations of where digital models will appear in the physical environment and real-time feedback if clipping occurs with any frames of the animation sequence.

3 Proposed System

Based on the design goals, we propose an MR accessible tool that enables users to author MR scenes intuitively while reducing workload, i.e., users employ a physical cube as a tangible proxy to place scaled-down models within a miniature model of the physical environment, with visual feedback to prevent clipping. Specifically, our system has three key features detailed as follows:

- **Authoring in miniature:** authoring scaled-down digital models within a miniature view of the environment. This bird's-eye perspective allows users to efficiently curate models across various locations, reducing manipulation time (i.e., searching, moving, and placing) compared to a first-person view. The miniature environment model is rendered in different colors according to semantic labels, providing contextual information to support model placement (DG1, 4).
- **Tangible interaction:** manipulating a physical cube as a proxy to place digital models. The familiar shape of the cube enables intuitive movement and rotation, simplifying digital model placement and accounting for collisions with the physical environment (DG2).
- **Immersive authoring with spatio-temporal bounding volumes:** authoring and verifying in situ, users can place models and instantly view authoring results in the physical environment, eliminating the need to go back-and-forth between desktop modeling and real-world testing. Model placement is augmented by collision detection between spatio-temporal bounding volumes and the scene, avoiding clipping of animation sequences without repeated testing (DG3, 4).

4 Implementation

We developed a prototype system using Unity, with Meta Quest 3 as the MR headset. The workflow of the system is shown in Fig. 1, and the implementation details are described below.

Miniature Modeling. We used the Mixed Reality Utility Kit (MRUK) to create the environment model, as shown in Fig. 1(a). For clarity of visualization, the environment model is rendered in wireframe, with vertex coloring applied based on semantic labels and bounding volumes extracted using MRUK.

Tangible Proxy. A printed physical cube with an edge length of 5.6 cm is used as a tangible 6-DoF controller to place digital models, tracked using OpenCV for Unity and Meta Quest's Passthrough Camera API, as shown in Fig. 2(a).

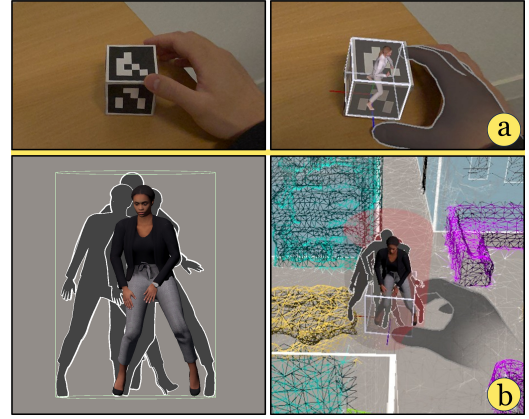


Figure 2: (a) Left: a physical cube used as a 6-DoF controller; Right: the cube is tracked as the user manipulates it for placing digital objects. (b) Left: visualization of *contributing frames* of an animated character; Right: our system flags potential collisions between the space-time bounds of the animated sequence and objects in the environment.

Spatio-temporal Bounding Volume. A bounding cylinder is used to contain the animation sequence of the character's vertices, with its radius defined by the minimum enclosing circle and its height by the range of the vertices' y -coordinates. When a collision is detected between the bounding cylinder and environment model, we present the *contributing frames* containing vertices on the cylinder boundary, rendered in gray with an outline to show potential clipping with the environment, as shown in Fig. 2(b).

5 Results and Conclusions

We tested the system in a furnished common area (80.87 m², initial capture within two minutes) with dynamic characters¹ and static furniture². The system allows easy placement of models and checking results without having to walk around the physical scene.

Overall, the work demonstrates the feasibility and advantages of combining WIM and tangible authoring of MR scenes with dynamic content, and provides technical insights into the implementation of systems that leverage such components.

In future work, we plan to assess and optimize system components such as the environment model quality, tracking of the tangible proxy, and precision of the spatio-temporal bounding volume. Most importantly, the developed prototype will serve as a test bed for user studies to evaluate the usability and task load of the proposed workflow compared to existing alternatives.

Acknowledgments

This research was supported by a Trinity College Dublin PhD Award, the Research Ireland Centre for Research Training in Artificial Intelligence (Grant No. 18/CRT/6223), and the Horizon Europe Framework Program under Grant Agreement 101070109.

¹3D human models, for illustrative purposes, were obtained from renderpeople.com and animated using Adobe Mixamo.

²3D furniture models by Nappin were obtained from the Unity Asset Store.

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